

Aahan Ghode

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SUMMARY

Computer Engineering student with experience in scientific computing, finite element analysis, and full-stack development, applying C++, Python, and MATLAB to simulation, data processing, and software development, with interests in embedded systems, software engineering, and hardware-software integration.

WORK EXPERIENCE

Simulation Engineer Jul 2026 — Aug 2026
Sunnybrook Research Institute Toronto, ON

- Modeled OnScale FEA simulations of radially-poled PZT-4 ring transducers across 3+ catheter geometries to evaluate acoustic beam focus
- Engineered MATLAB pipelines to parse binary time-history data, reducing simulation post-processing time by 60%
- Mapped spatial pressure distributions to characterize acoustic energy delivery and cavitation behavior
- Calculated piezoelectric mode purity from displacement fields to verify clean length-mode excitation
- Analyzed dynamic mechanical stress distributions to identify peak standing-wave stress and establish safe drive voltage limits

Lead Assistant Teacher Jun 2023 — Jun 2026
Spirit of Math Schools Inc. Oakville, ON

- Taught advanced mathematics curricula to 40+ Grade 9–12 students
- Mentored 15+ students weekly on advanced coursework and problem-solving strategies
- Developed and delivered lessons focused on mathematical reasoning, competition-style problem solving, and advanced concepts
- Participated in weekly administrative reviews with senior faculty to evaluate student progress and improve instruction
- Co-developed personalized academic learning plans with lead instructors based on individual student needs

Full Stack Developer Intern Nov 2024 — May 2025
Vertex Dimension Remote

- Developed custom front-end components and integrated third-party Shopify apps using HTML/CSS across 3 client sites
- Supported 2 project managers across 10+ client technical meetings, translating business requirements into engineering specifications
- Optimized site navigation and SEO structures, improving site engagement by 35%
- Collaborated with developers and project managers to troubleshoot implementation issues and deliver client requirements

Database Administrator Intern Aug 2023 — Oct 2023
RS Tec Systems Inc. Oakville, ON

- Engineered an automated resume parser capable of batch processing 200+ applicant documents
- Parsed and filtered unstructured resume data across 3 file formats (.pdf, .docx, .doc), reducing manual screening time by 75%
- Collaborated with 2 senior administrators to review feedback and implement security enhancements
- Integrated Google Sheets API to automate spreadsheet synchronization and eliminate manual data entry
- Designed a responsive Flask and Bootstrap UI to streamline database management and administrative record entry

SKILLS

Languages: C, C++, Python, MATLAB, Java, JavaScript
Engineering & Technical: OnScale, Finite Element Analysis, Data Structures and Algorithms
Frameworks & APIs: React, Next.js, Flask, Django, Node.js, REST APIs
Tools: Linux, Git, GitHub, OnScale, Google Sheets API
Spoken Languages: English, Marathi, French

HONOURS AND AWARDS

Jun. 2026	Minister’s Certificate of Recognition for Community Involvement (200+ Hours)	Ontario Ministry of Education
May. 2026	Schulich Leader Scholarship (\$120K)	University of Toronto
Nov. 2025	Certificate of Distinction (Top 25%) CSMC	University of Waterloo
Apr. 2025	Certificate of Distinction (Top 25%), Hypatia Math Contest	University of Waterloo
Jul. 2024	AP Scholar (Scored 3 or higher on 3 or more AP Exams)	College Board

EDUCATION

University of Toronto <i>Bachelor of Applied Science, Computer Engineering</i>	Toronto, ON Sept 2026 — Apr 2031
Iroquois Ridge High School <i>Ontario Secondary School Diploma, Clustered Program</i>	Oakville, ON Sept 2022 — Jun 2026